

Mamba2: A Linear Attention Layer

Mamba2 [1] is an alternative information propagation algorithm belonging to the steadily growing class of *linear* $O(\text{seqlen})$ attention mechanisms. Its GPU-aware design enables competitive hardware utilization to quadratic attention during training, and reduces the time and space complexity during decode from $O(\text{seqlen})$ to $O(1)$.

'Columbidae is a bird family consisting of doves and pigeons. It is the only family in the order Columbiformes.' [2]



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Domain	Eukaryota
Kingdom	Animalia
Phylum	Chordata
Class	Aves
Clade	Columbimorphae
Order	Columbiformes
Family	Columbidae
Type genus	Columba

This box is styled differently compared to the others. To make such changes persistent across the whole poster, we can use these functions:

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#pop.update-poster-layout(...)
#pop.update-theme()
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You can find more information on the documentation site under jonaspleyer.github.io/peace-of-posters/ .



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General Relativity

Einstein’s brilliant theory of general relativity starts with the field equations [4].

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

However, they have nothing to do with doves.

Peace be with you



Figure 3: ‘Doves [...] are used in many settings as symbols of peace, freedom or love. Doves appear in the symbolism of Judaism, Christianity, Islam and paganism, and of both military and pacifist groups.’ [5].

Etymology

Pigeon is a French word that derives from the Latin pīpiō, for a ‘peeping’ chick, while dove is an ultimately Germanic word, possibly referring to the bird’s diving flight. The English dialectal word culver appears to derive from Latin columba [6]. A group of doves is called a “dule”, taken from the French word deuil (‘mourning’) [7].

References

[1] T. Dao and A. Gu, “Transformers are SSMS: Generalized Models and Efficient Algorithms Through Structured State Space Duality.” [Online]. Available: <https://arxiv.org/abs/2405.21060>

[2] Wikipedia, “Columbidae — Wikipedia, The Free Encyclopedia.” 2024.

[3] Wikipedia, “File:Treron vernans male - Kent Ridge Park.jpg — Wikipedia, The Free Encyclopedia.” 2024.

[4] A. Einstein, “Die Grundlage der allgemeinen Relativitätstheorie [AdP 49, 769 (1916)],” *Annalen der Physik*, vol. 517, no. S1, pp. 517–571, 1916, doi: 10.1002/andp.2005517s151.

[5] Wikipedia, “Doves as symbols — Wikipedia, The Free Encyclopedia.” 2024.

[6] Wikipedia, “Online Etymology Dictionary — Wikipedia, The Free Encyclopedia.” 2024.

[7] J. Lipton, *An Lipton James : Exaltation of larks*. Harlow, England: Penguin Books, 1991.

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